

Assignment - 4

Course: MST501-1: Python
Programming for Statistics



Done by:

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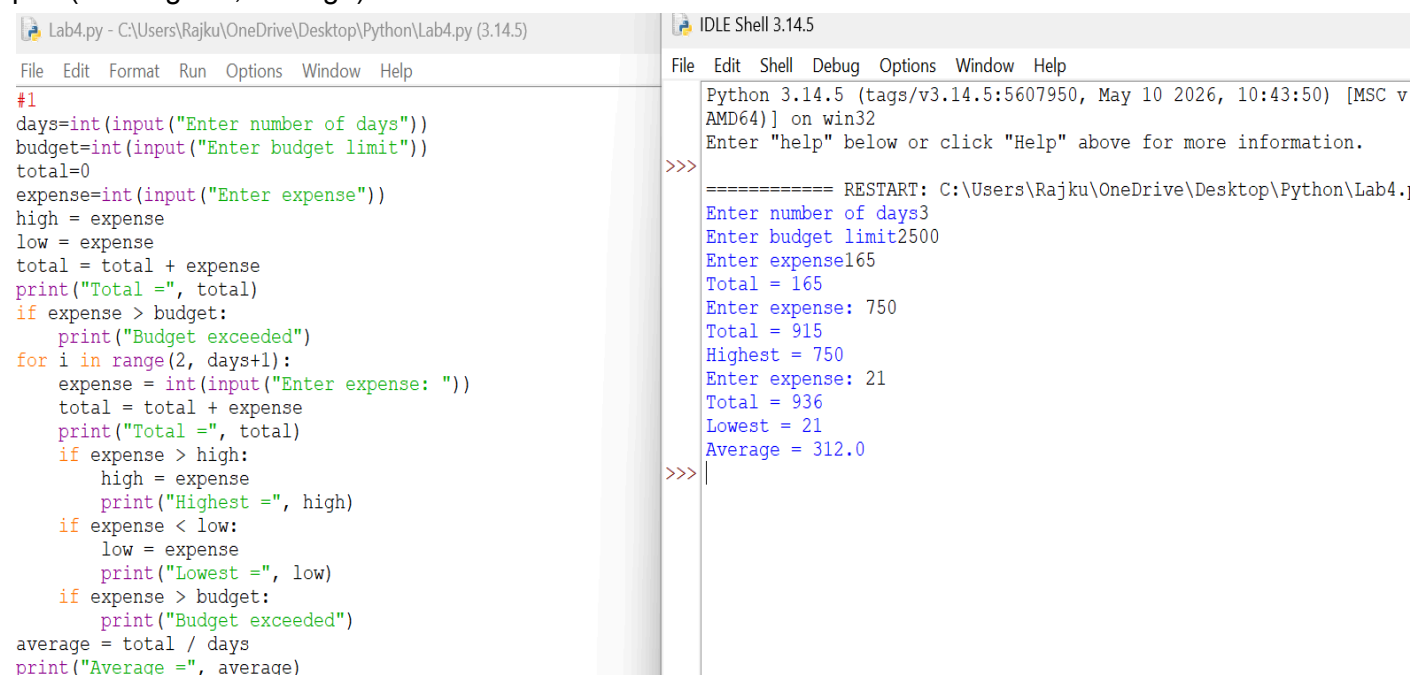
Register Number: 2648103

Date: 25/06/2026

1. Develop a Python program that records a user's daily expenses for an entire month. Using loops, calculate total spending, average daily expenditure, highest and lowest expenses, and identify days where spending exceeded a user-defined budget limit. Generate a monthly financial summary.

Ans-

```
days=int(input("Enter number of days"))
budget=int(input("Enter budget limit"))
total=0
expense=int(input("Enter expense"))
high = expense
low = expense
total = total + expense
print("Total =", total)
if expense > budget:
    print("Budget exceeded")
for i in range(2, days+1):
    expense = int(input("Enter expense"))
    total = total + expense
    print("Total =", total)
    if expense > high:
        high = expense
        print("Highest =", high)
    if expense < low:
        low = expense
        print("Lowest =", low)
    if expense > budget:
        print("Budget exceeded")
average = total / days
print("Average =", average)
```



The screenshot displays a Python IDE with two panels. The left panel shows the source code for a file named 'Lab4.py'. The code prompts the user for the number of days, budget limit, and daily expenses, then calculates the total, average, highest, and lowest expenses. The right panel shows the IDLE Shell output, which includes the program's execution flow and the user's input values.

```
Lab4.py - C:\Users\Rajku\OneDrive\Desktop\Python\Lab4.py (3.14.5)
File Edit Format Run Options Window Help
#1
days=int(input("Enter number of days"))
budget=int(input("Enter budget limit"))
total=0
expense=int(input("Enter expense"))
high = expense
low = expense
total = total + expense
print("Total =", total)
if expense > budget:
    print("Budget exceeded")
for i in range(2, days+1):
    expense = int(input("Enter expense: "))
    total = total + expense
    print("Total =", total)
    if expense > high:
        high = expense
        print("Highest =", high)
    if expense < low:
        low = expense
        print("Lowest =", low)
    if expense > budget:
        print("Budget exceeded")
average = total / days
print("Average =", average)
```

```
IDLE Shell 3.14.5
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v
AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Lab4.
Enter number of days3
Enter budget limit2500
Enter expense165
Total = 165
Enter expense: 750
Total = 915
Highest = 750
Enter expense: 21
Total = 936
Lowest = 21
Average = 312.0
>>>
```

2. Create a Python program that simulates a parking facility for an entire day. Using loops, process vehicle entries and exits, calculate parking charges based on duration, track occupancy levels, identify peak usage periods, and generate end-of-day statistics such as total revenue and average parking time.

Ans- vehicles = int(input("Enter number of vehicles"))

tr = 0

tt= 0

peak=0

occupancy=0

for i in range(1, vehicles+1):

time = int(input("Enter parking time (hours)"))

charge = time*20

tr = tr + charge

print("Total Revenue =", tr)

tt = tt + time

occupancy = occupancy + 1

if occupancy > peak:

peak = occupancy

print("Peak Occupancy =", peak)

average = tt / vehicles

print("Average Parking Time =", average)

<pre>File Edit Format Run Options Window Help #2 vehicles = int(input("Enter number of vehicles")) tr = 0 tt= 0 peak=0 occupancy=0 for i in range(1, vehicles+1): time = int(input("Enter parking time (hours)")) charge = time*20 tr = tr + charge print("Total Revenue =", tr) tt = tt + time occupancy = occupancy + 1 if occupancy > peak: peak = occupancy print("Peak Occupancy =", peak) average = tt / vehicles print("Average Parking Time =", average)</pre>	<pre>File Edit Shell Debug Options Window Help Python 3.14.5 (tags/v3.14.5:5607950, May 10 2026, 10:43:50) [MSC v.1944 64 bit (AMD64)] on win32 Enter "help" below or click "Help" above for more information. >>> ===== RESTART: C:\Users\Rajku\OneDrive\Desktop\Python\Lab4.py ===== Enter number of vehicles4 Enter parking time (hours)12 Total Revenue = 240 Peak Occupancy = 1 Enter parking time (hours)25 Total Revenue = 740 Peak Occupancy = 2 Enter parking time (hours)12 Total Revenue = 980 Peak Occupancy = 3 Enter parking time (hours)5 Total Revenue = 1080 Peak Occupancy = 4 Average Parking Time = 13.5 >>></pre>
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3. Design a Python program that manages sales transactions for an online store over multiple days. Using nested loops, process customer orders, update inventory levels, calculate daily and cumulative revenue, identify best-selling products, detect low-stock items, and generate a comprehensive business performance report. The system should continue operating until the administrator chooses to terminate it.

```
Ans-revenue=0
stock=50
sold=0
choice="yes"
while choice=="yes":
    orders= int(input("Enter number of orders for the day"))
    daily= 0
    for i in range(1, orders+1):
        quantity=int(input("Enter quantity sold "))
        stock = stock - quantity
        print("Stock Left =", stock)
        sold=sold + quantity
        print("Items Sold =", sold)
        amount=quantity * 100
        daily=daily + amount
        print("Daily Revenue =", daily)
        revenue=revenue + amount
        print("Total Revenue =", revenue)
        if stock < 10:
            print("Low Stock Warning")
    choice=input("Choice")
```

The screenshot shows a Python IDE with two panes. The left pane displays the Python code for managing sales transactions. The right pane shows the execution output, which includes the program's initialization, a loop for processing orders, and a low stock warning.

```
File Edit Format Run Options Window Help
#3
revenue=0
stock=50
sold=0
choice="yes"
while choice=="yes":
    orders= int(input("Enter number of orders for the day"))
    daily= 0
    for i in range(1, orders+1):
        quantity=int(input("Enter quantity sold: "))
        stock = stock - quantity
        print("Stock Left =", stock)
        sold=sold + quantity
        print("Items Sold =", sold)
        amount=quantity * 100
        daily=daily + amount
        print("Daily Revenue =", daily)
        revenue=revenue + amount
        print("Total Revenue =", revenue)
        if stock < 10:
            print("Low Stock Warning")
    choice=input("Choice")
```

```
File Edit Shell Debug Options Window Help
Python 3.14.5 (tags/v3.14.5:5607950, May 10
AMD64)] on win32
Enter "help" below or click "Help" above for
>>>
===== RESTART: C:\Users\Rajku\OneDrive\
Enter number of orders for the day3
Enter quantity sold: 25
Stock Left = 25
Items Sold = 25
Daily Revenue = 2500
Total Revenue = 2500
Enter quantity sold: 25
Stock Left = 0
Items Sold = 50
Daily Revenue = 5000
Total Revenue = 5000
Low Stock Warning
Enter quantity sold: 5
Stock Left = -5
Items Sold = 55
Daily Revenue = 5500
Total Revenue = 5500
Low Stock Warning
ChoiceNo
>>>
```

